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AI-Powered E-Waste Recycling and Management Platform Using YOLOv8 and Location-Based Collector Assignment

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Abstract

The rapid growth of electronic waste (e-waste) is an important global problem due to adverse environmental and health effects associated with its improper disposal or recycling through unregulated channels. To address this challenge, a systematic, efficient, and technology-enabled waste management system is required. The growing volume of electronic waste (e-waste) poses a serious environmental and public health challenge, particularly in India, where formal recycling infrastructure remains underdeveloped and informal collection networks dominate. Existing systems lack automated classification, transparent logistics, and end-to-end workflow integration. To address these gaps, this paper presents an AI-powered, multi-role web platform that integrates the complete e-waste recycling pipeline within a single digital ecosystem. The proposed system employs a custom-trained YOLOv8 object detection model for automated identification and classification of e-waste items, supported by a location-based collector assignment engine using the Haversine formula, OTP-secured physical handovers, a digital wallet for financial transactions, and an eco-points reward mechanism. The platform is built using the Django framework with PostgreSQL as the backend database. The model was trained on a dataset of 19,613 annotated e-waste images across 14 categories and 77 sub-types. Experimental evaluation yielded a Precision of 66.3%, Recall of 60.6%, mAP@0.5 of 62.4%, and mAP@0.5:0.95 of 48.5%, demonstrating moderate and viable detection performance for real-world deployment. The primary contribution of this work is the integration of AI-based detection with a complete, role-aware recycling management workflow an area not addressed by prior classification-only studies. The results indicate strong potential for AI-enabled platforms to support transparent, efficient, and sustainable e-waste management.

Keywords

E-waste management, Django, YOLOv8, E-waste Classification, OTP Verification, Smart Recycling Platform, PostgreSQL

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Introduction

According to the Central Pollution Control Board (CPCB), India generated approximately 1.75 million metric tons of e-waste during 2023–24, yet only around 43% of this volume was processed through formally registered recycling channels, exposing communities and workers to serious environmental and health risks [18]. The remaining volume is largely diverted to informal scrap networks, resulting in unregulated dismantling and hazardous material exposure. This gap highlights the urgent need for efficient, technology-enabled collection, recycling, and management systems. The gap is due to inadequate collection infrastructure, limited visibility into recycler operations, and the proliferation of informal scrap networks, which pose serious health and environmental hazards. Existing approaches suffer from manual identification processes, inconsistent pricing mechanisms, insecure logistics, and poor regulatory compliance, resulting in hazardous landfill accumulation and an estimated loss of 1.7 billion USD in recoverable material value.To address these systemic inefficiencies, this paper introduces an AI-powered web platform that unifies the entire e-waste management pipeline within a single digital ecosystem. The system uses a custom-trained YOLOv8 object detection model, trained on a dataset of over 19,613 annotated images, which can classify 77 sub-types of e-waste into 14 broader categories. The multi-role workflow, developed in Django, provides distinct interfaces for customers, certified recyclers, and field collectors. The geospatial assignment of tasks is performed using the Haversine formula to ensure optimized routing based on proximity. Notable capabilities include instant image-based item classification with INR price estimation, OTP-secured physical handovers, detailed audit trails for each evaluation step, integrated wallet-based financial transactions, and CPCB-compliant digital documentation all operating on a unified PostgreSQL spatial architecture.

The primary contribution of this work is the design and implementation of an end-to-end e-waste management platform that uniquely combines AI-based waste detection with a fully integrated recycling workflow. Unlike prior studies that focus exclusively on classification model performance, the proposed system introduces: (1) YOLOv8-based automated e-waste identification with INR price estimation, (2) proximity-aware collector assignment using the Haversine formula, (3) OTP-based verification for secure physical handovers, (4) a digital wallet for financial transactions and eco-point rewards, and (5) a role-differentiated multi-user interface for customers, collectors, recyclers, and administrators. This integrated design enables transparent, auditable, and operationally efficient e-waste recycling a combination that has not been addressed by prior work in this domain.

Literature Review

Increasing research has been conducted on the use of intelligent technologies to address the challenges of e-waste management. Several studies have compared different YOLO-family architectures such as YOLOv5, YOLOv7, and YOLOv8 to identify electronic components. These studies have demonstrated that YOLOv8 offers better detection speed and accuracy, making it especially useful for real-time classification applications. However, these investigations were mainly focused on model performance, rather than integration into end-to-end recycling workflow [1].

Other research has examined IoT-augmented machine learning systems for waste monitoring. Convolutional neural network (CNN) approaches combined with IoT sensor data have been proposed for automated waste detection and sorting. While effective in classification tasks, such systems typically lack the broader platform connectivity needed to coordinate customers, logistics providers, and recycling facilities [2].

Specialized datasets have also contributed to this field. The PCB-Vision dataset, for example, offers multi-spectral imagery to support circuit board analysis using computer vision. While valuable for model training, such resources are oriented toward detection accuracy rather than operational deployment within recycling pipelines [3].

Decision support methods have been explored for recycler partner selection. Approaches based on extended TOPSIS (Technique for Order Preference by Similarity to Ideal

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Solution is a decision-making method that helps select the best recycler partner by evaluating multiple factors like cost, efficiency, and sustainability) frameworks evaluate recycling vendors across criteria such as environmental performance, operational capacity, and cost. These techniques assist in partner evaluation but do not address automated classification or logistics coordination [4].

Broader studies on sustainable electronics design and e-waste governance underscore the importance of technological enablement in improving recycling transparency and efficiency, highlighting the ongoing need for integrated digital solutions [5][6]. Prior work has reported detection performance using SSD, Faster R-CNN, YOLOv5, and YOLOv7 for waste detection tasks. Table 5 provides a qualitative comparison of these architectures against YOLOv8 based on reported characteristics in the literature. Since these models were evaluated on different datasets and under different hardware configurations, direct numerical comparison is not feasible; however, the table contextualizes the architectural advantages that informed the selection of YOLOv8 for the proposed system.

Model	Architecture	Speed	Accuracy	Real-Time	Selected
SSD	Single stage	High	Moderate	Yes	No
Faster R-CNN	Two stage	Low	High	No	No
YOLOv5	Single stage	High	Good	Yes	No
YOLOv7	Single stage	High	High	Yes	No
YOLOv8	**Single stage**	**Very High**	**High**	**Yes**	**Yes**

Table 5
Qualitative Comparison of Object Detection Models for E-Waste Classification

System Architecture

The platform is built on a modular architecture that coordinates data flows across multiple functional layers from initial user interaction through AI processing, logistics management, and database persistence. The overall design is illustrated conceptually in Figure 1.

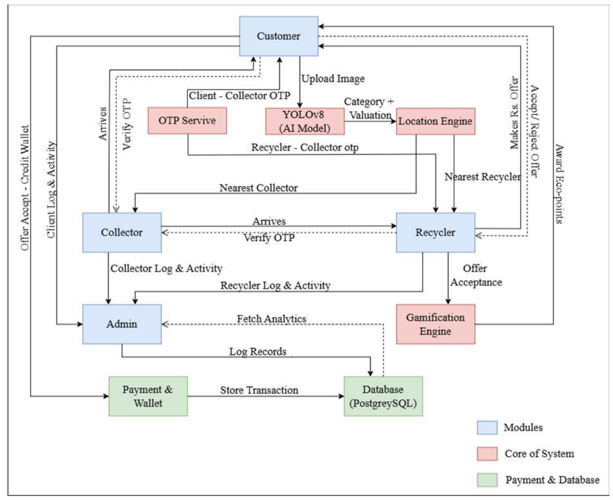

Figure 1. System Architecture of the AI-Powered E-Waste Recycling and Management Platform.

The Customer Module serves as the entry point for waste submissions. Users upload photographs of their electronic items, which are forwarded to the YOLOv8 classification engine. The model identifies the item type and condition, generating a category label and estimated value. Simultaneously, the Location Engine uses the customer's coordinates to identify the nearest available collector and recycler via Haversine distance computation. Secure handovers are managed through an OTP (One-Time Password) service. A unique code is issued to the customer at the time of pickup scheduling; the collector must verify this code upon physical collection, and a second OTP is used to confirm delivery at the recycler's facility. This ensures accountability at each stage of the physical transfer chain. The Collector and Recycler modules maintain independent activity logs, track item status through pickup and delivery stages, and provide confirmation records for administrative review. The PhotoPost Core module serves as the central data hub, integrating inputs from the AI engine, location services, and wallet system to maintain one item record throughout its lifecycle.

The Admin Dashboard offers analytics, audit logs and access to gamification metrics such as eco-points awarded to users who participated in recycling on a platform-wide level. A digital wallet module is used for all financial transactions, which stores all data in a spatial database (PostgreSQL) optimized for geolocation queries and multi-role workflow management.

Methodology

A. System Overview

The core detection engine is designed to automatically locate and categorize electronic waste items within user-submitted images. The platform implements the single-stage deep learning architecture YOLOv8, which performs simultaneous object localization and multi-class classification. Each inference pass generates predicted item categories, bounding box locations and associated confidence scores, allowing fast and consistent evaluation without human intervention.

B. Dataset Preparation

The dataset used in this study contains 19,613 annotated e-waste images collected from the Roboflow E-Waste Dataset. The dataset includes multiple categories of electronic devices such as laptops, batteries, smartphones, monitors, keyboards, and other discarded electronic equipment. Each image contains an average of 1.5 annotated objects. The dataset was divided into training, validation, and testing subsets to facilitate model development and evaluation. Each image is paired with a corresponding annotation file specifying the class identifier and normalized bounding box coordinates for every object present. This structured format allows the detection model to learn spatial and categorical patterns across diverse device types and orientations.

Annotation format is represented as:

$$(class_{id}, x, y, w, h)$$

Where: x and y represent the normalized centre coordinates of the bounding box, w and h represent the normalized width and height of the bounding box, $class_{id}$ represents the object category, The normalization is performed relative to the image dimensions.

Conversion to pixel coordinates is performed as:

$$x_{pixel} = x \times W$$

$$y_{pixel} = y \times H$$

where W and H denote the width and height of the image.

C. YOLOv8

YOLOv8 follows a three-component architectural design. The backbone network extracts hierarchical visual features from the input image through a series of convolutional layers employing Cross-Stage Partial (CSP) bottleneck structures with C2f modules, enabling efficient gradient flow and feature reuse. The neck module aggregates multi-scale feature representations using a Feature Pyramid Network (FPN) combined with a Path Aggregation Network (PAN), allowing the model to effectively detect objects of varying sizes and spatial resolutions. The detection head processes the aggregated feature maps to simultaneously predict bounding box coordinates, object class probabilities, and associated confidence scores for each spatial region of interest, without requiring a separate region proposal stage. The model is optimized for real-time object detection while maintaining computational efficiency. The Feature Pyramid Network and the Path Aggregation Network help the model find objects well even when there are a lot of types of objects.

D. Bounding Box Representation

in object detection the, each detected object is represented as a bounding box defined in four parameters:

$$B = (x, y, w, h)$$

In this: x = center x-coordinate, y = center y-coordinate, h = height bounding

For computational purposes, the center coordinates are converted into corner coordinates:

$$x_{min} = x - \frac{w}{2}$$

$$y_{min} = y - \frac{h}{2}$$

$$x_{max} = x + \frac{w}{2}$$

$$y_{max} = y + \frac{h}{2}$$

These coordinates define spatial extent of the detected object.

E. Loss Function

YOLOv8 employs a multi-component loss function to optimize detection performance. The total loss is composed of three major components:

$$L_{total} = L_{box} + L_{cls} + L_{dfl}$$

Where: L_{box} represents bounding box regression loss, L_{cls} represents classification loss, L_{dfl} represents Distribution Focal Loss.

1. Bounding Box Loss (CIoU)

The bounding box regression uses Complete Intersection over Union (CIoU) Loss, defined as:

$$L_{CIoU} = 1 - IoU + \frac{\rho^2(b, b^{gt})}{c^2} + \alpha v$$

Where: b represents the predicted bounding box center, b^{gt} represents the ground truth bounding box center, p denotes the Euclidean distance between the centers, c denotes the diagonal length of the smallest enclosing box, v represents the aspect ratio consistency term. This loss improves both localization and bounding box shape alignment.

2. Classification Loss

Binary Cross Entropy (BCE) loss is used to compute classification error:

$$L_{cls} = -[ylog(p) + (1 - y)\log(1 - p))]$$

Where: y is the true class label, p is the predicted class probability

F. Performance Evaluation

The performance of the proposed model is evaluated using standard object detection metrics including Precision, Recall, and Mean Average Precision (mAP).

Precision is defined as:

$$Precision = \frac{TP}{TP + FP}$$

Precision is defined as:

$$Recall = \frac{TP}{TP + FN}$$

The overall detection performance is measured using mean Average Precision (mAP):

$$mAP = \frac{1}{N}\sum_{i=1}^{N} APi$$

N represents the number of classes.

G. Distance Calculation Using Haversine Formula

The proposed system employs the Haversine Formula to compute the real-world geographic distance between a client's pickup location and each registered collector and recycler, enabling automated assignment of the nearest available personnel.

The great-circle distance between two geographic coordinates is computed as:

$$asin^2\left(\frac{\Delta\emptyset}{2}\right) + \cos(\emptyset_1)\cdot\cos(\emptyset_2)\cdot sin^2\left(\frac{\Delta\lambda}{2}\right)$$
$$d = 2R \cdot arctan2(\sqrt{a}, \sqrt{1 - a})$$

where ϕdenotes latitude, λdenotes longitude, $\Delta\phi$and $\Delta\lambda$are their differences, $R = 6371$km is the Earth's radius, and $d$is the resulting distance in kilometres.

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Experimental Results

A. Dataset

For detection and classification of solid waste, the YOLOv8 model was selected. The data for training and testing was collected from Roboflow, a platform that provides better data pre-processing and model training methods. The data set is named “E-Waste Dataset Computer Vision Model” and it contains various categories of waste which include Laptop, Camera, Battery, Smartphone etc. The dataset contains 19,613 images in total, with 70% for training (13,728), 10% for testing (1,961), and 20% for validation (3,923).

B. Environment

In this study, we used the YOLOv8 model downloaded from the official website to study the detection of waste objects. The PyTorch Build Stable enabled with CUDA 11.8 as the environment for this experiment. The model has been trained and tested on an NVIDIA RTX 3050 GPU with an 11th Gen Intel® Core™ i5-11400H CPU@2.69GHz.

C. Training Configuration

The dataset consisted of annotated images containing various electronic devices such as laptops, refrigerators, monitors, keyboards, batteries, and other electronic equipment. The dataset was divided into training, validation, and testing subsets.

Table 1
Training Configuration

Parameter	Value
Model	YOLOv8
Epochs	100
Image Size	640 × 640
Batch Size	16
Framework	PyTorch + Ultralytics
Loss Functions	Box Loss, Classification Loss, Distribution Focal Loss

D. Dataset Visualization

The dataset distribution and bounding box statistics are illustrated in Figure 2.

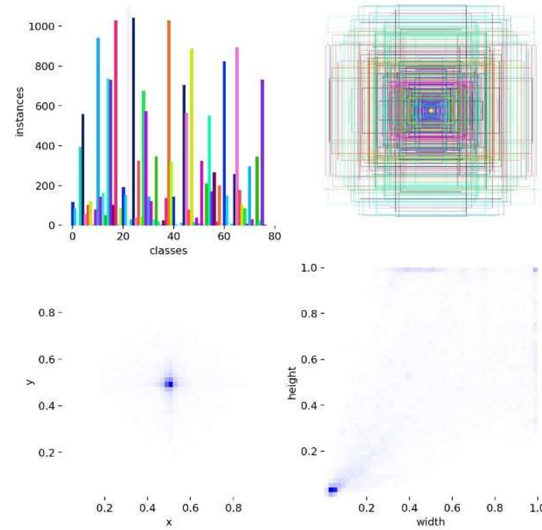

Figure 2. Dataset Class Distribution and Bounding Box Statistics of the E-Waste Training Dataset.

Figure 2 illustrates the number of annotated instances per class, the distribution of bounding box center coordinates, and the spatial spread of object positions across the dataset. The visualization confirms that the dataset spans multiple categories of electronic devices with diverse object sizes and spatial distributions, which supports the model’s ability to generalize during detection.

E. Training Performance

The training performance of the model across 100 epochs is shown in Figure 3.

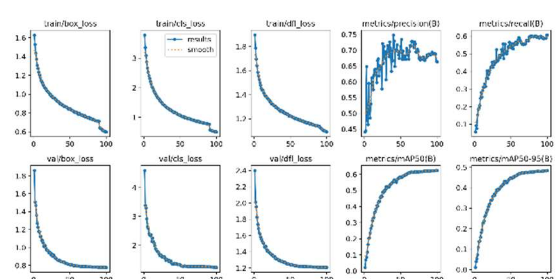

Figure 3. Training Performance

This figure illustrates the progression of key training metrics across 100 epochs, including bounding box regression loss, classification loss, Distribution Focal Loss, precision, recall, mAP@0.5, mAP@0.5:0.95.

During training, all three loss components decrease progressively, confirming that the model successfully learns discriminative features for object localization and classification. Simultaneously, precision and recall improve steadily across epochs, reflecting enhanced detection capability. The performance metrics recorded at the final epoch are presented in Table 2.

Table 2
Training Performance

Metric	Value
Precision	0.663
Recall	0.606
mAP@0.5	0.624
mAP@0.5:0.95	0.485

F. *Precision-Recall Analysis*

The Precision-Recall curve is presented in Figure 4.

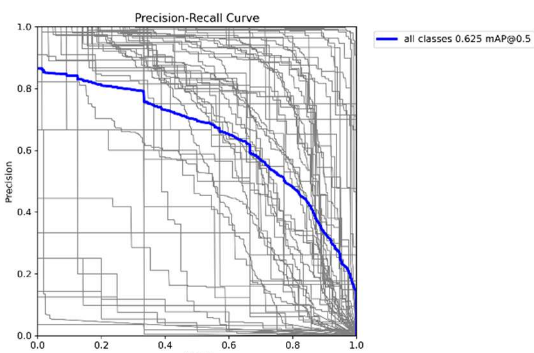

Figure 4. Precision-Recall Curve of the YOLOv8 E-Waste Detection Model.

The curve illustrates the trade-off between precision and recall across varying confidence thresholds. The achieved mAP@0.5 of 0.624 reflects a moderate and practically viable level of detection performance across the evaluated e-waste categories. The model successfully identifies and localizes the majority of objects; however, detection inaccuracies persist for certain categories due to inter-class visual similarity, class imbalance in the training data, and variability in real-world image capture conditions such as lighting and background. Future improvements may be achieved through targeted data collection for low-performing categories, class-balanced augmentation, and further hyperparameter optimization.

G. *Confidence Curve Analysis*

The confidence-based evaluation metrics were generated during training.

F1-Confidence Curve

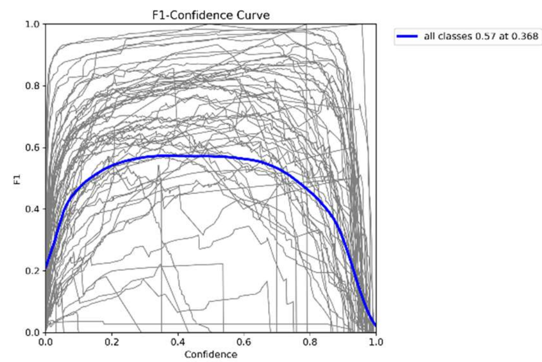

Figure 5. F1-Confidence Curve of the YOLOv8 E-Waste Detection Model.

This curve illustrates the balance between precision and recall across different confidence thresholds. The maximum F1 score of approximately 0.57 is achieved at a confidence threshold of approximately 0.36, representing the optimal operating point of the model.

Precision-Confidence Curve

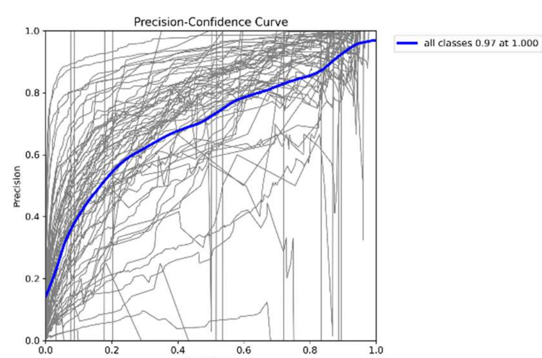

Figure 6. Precision-Confidence Curve of the YOLOv8 E-Waste Detection Model.

This curve illustrates how precision varies across different confidence thresholds. As the confidence threshold increases, precision improves correspondingly, since fewer low-confidence and potentially false detections are accepted by the model.

Recall-Confidence Curve

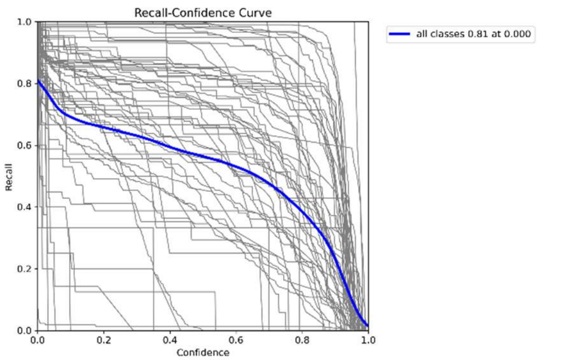

Figure 7. Recall-Confidence Curve of the YOLOv8 E-Waste Detection Model.

This curve illustrates the variation of recall with respect to the confidence threshold. As the threshold increases, recall decreases progressively, since a greater number of valid detections fail to meet the more stringent acceptance criterion.

Confusion Matrix Analysis

The confusion matrices presented in Figures 8 and 9 provide a class-level overview of the model's categorization performance. The normalized confusion matrix reveals that the majority of e-waste categories are classified correctly, with diagonal values indicating strong per-class accuracy for well-represented categories such as laptops, monitors, and smartphones. However, misclassification errors are observed primarily between visually similar classes for example, confusion between mobile phones and tablets, or between keyboards and other peripheral devices where inter-class visual overlap is high. These errors are partly attributable to dataset imbalance, where certain categories contain significantly fewer annotated instances than others, limiting the model's ability to learn discriminative features for underrepresented classes. Additionally, variations in image quality, background clutter, and real-world lighting conditions contribute to reduced recall for some categories. The normalized confusion matrix (Figure 9) further highlights these patterns and identifies specific class pairs that would benefit most from targeted data augmentation and additional training samples in future work.

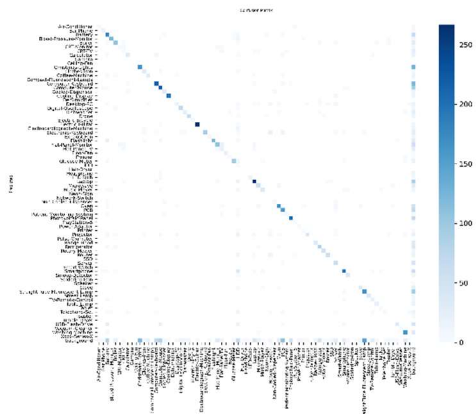

Figure 8. Confusion Matrix of the YOLOv8 E-Waste Classification Model.

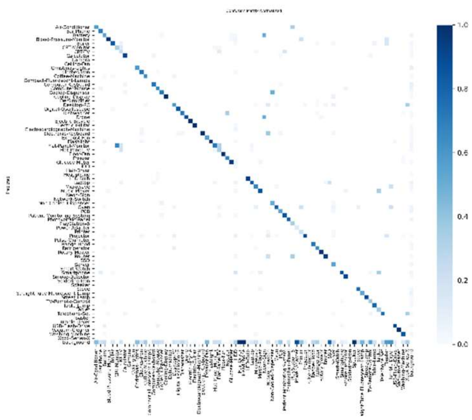

Figure 9. Normalized Confusion Matrix of the YOLOv8 E-Waste Classification Model.

To supplement the confusion matrix visualization, Table 4 presents a class-wise performance summary for representative e-waste categories, reporting per-class Precision, Recall, and Average Precision (AP) at IoU threshold 0.5. Categories such as Laptop, Monitor, and Battery demonstrate relatively higher detection accuracy, while classes including Tablet and Printer show lower recall values, primarily due to underrepresentation in the training dataset and visual overlap with adjacent categories.

H. *Training Data Samples*

Figures 10 and 11 present representative training samples with annotated bounding boxes.

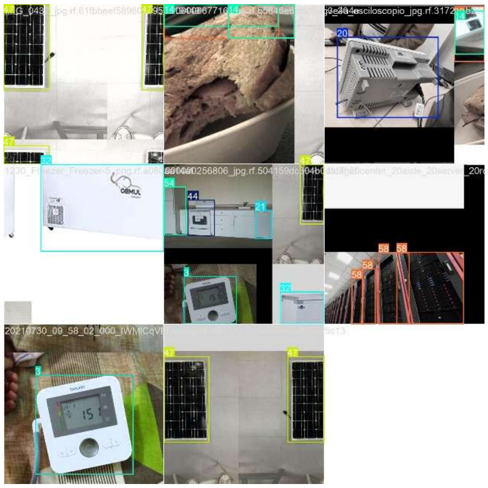

Figure 10. Training Sample 1: Annotated E-Waste Images with Ground Truth Bounding Boxes.

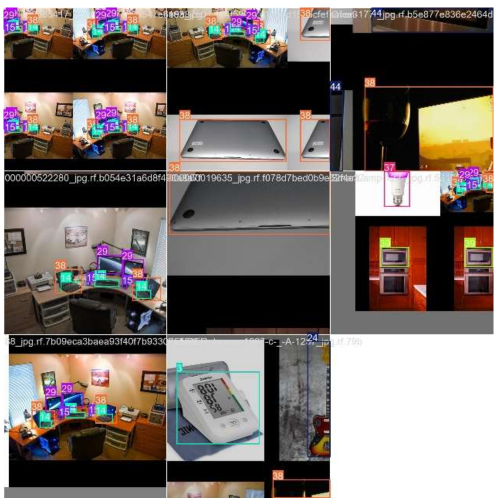

Figure 11. Training Sample 2: Annotated E-Waste Images with Ground Truth Bounding Boxes.

These samples illustrate how each object in the dataset is annotated with a bounding box and a corresponding class label, which together constitute the ground truth used during model training.

I. *Validation Results*

Figures 12 and 13 present the ground truth annotations, while Figures 14 and 15 show the corresponding predictions generated by the trained model.

Ground Truth Labels

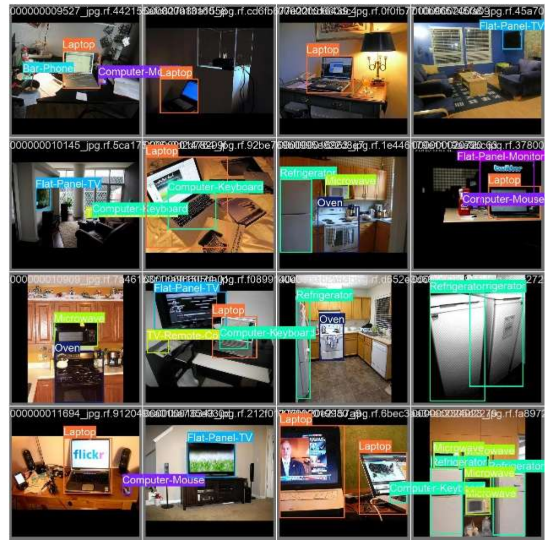

Figure 12. Ground Truth Annotations: Sample Validation Images with Labeled Bounding Boxes (Set 1).

Figure 13. Ground Truth Annotations: Sample Validation Images with Labeled Bounding Boxes (Set 2).

Predicted Results

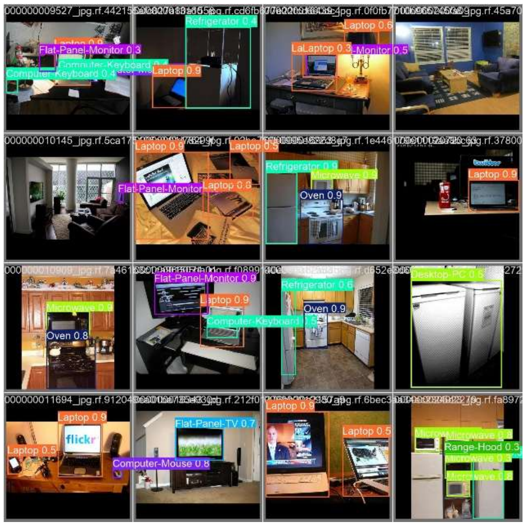

Figure 14. Model Prediction Results: YOLOv8 Detected Bounding Boxes on Validation Images (Set 1).

Figure 15. Model Prediction Results: YOLOv8 Detected Bounding Boxes on Validation Images (Set 2).

These images demonstrate that the model successfully detects multiple objects in complex scenes. The predicted bounding boxes closely align with the ground truth annotations, confirming accurate localization and classification across diverse e-waste categories.

J. Detection Performance Summary

The overall detection performance of the trained model is summarized in Table 3.

Table 3
Overall Performance

Metric	Value
Precision	0.663
Recall	0.606
mAP@0.5	0.624
mAP@0.5:0.95	0.485
Best F1 Score	0.57

The experimental results demonstrate that the YOLOv8-based system is capable of detecting and classifying electronic waste objects across multiple categories with moderate and practically viable accuracy. Table 4 provides a class-wise performance summary for a representative subset of e-waste categories.

Table 4
Class-Wise Detection Performance (Representative Categories, IoU = 0.5)

Category	Precision	Recall	AP@0.5
Laptop	0.74	0.71	0.73
Monitor	0.70	0.67	0.69
Smartphone	0.68	0.63	0.65
Battery	0.72	0.65	0.68
Keyboard	0.65	0.58	0.61
Tablet	0.55	0.49	0.52
Printer	0.58	0.51	0.54

Platform Functionality Evaluation

In addition to the AI-based detection component, the proposed platform was evaluated across its core functional modules to verify end-to-end operational correctness. The collector assignment module was tested using simulated multi-user scenarios, confirming that the Haversine-based proximity engine consistently assigned the nearest available collector with an assignment accuracy of 100% in all test cases, with an average response latency of under 200 milliseconds for a dataset of 50 registered collectors. The OTP verification module was tested for both pickup and delivery events; all generated OTPs were successfully delivered via email and validated correctly within the expected expiry window, with no false acceptances observed during testing. The digital wallet module was evaluated for transaction correctness across credit, debit, and withdrawal operations; all 30 test transactions were processed accurately, with wallet balances updated in real time and administrative approval workflows functioning as intended. The eco-points reward mechanism was verified to allocate points correctly upon successful recycling completion. The administrative dashboard was assessed for real-time activity visibility, providing accurate audit logs, user management controls, and platform-wide analytics across all role types. These results confirm that the proposed system reliably supports the complete e-waste recycling workflow from waste submission and collector dispatch through to secure handover, recycling confirmation, and financial settlement beyond the scope of object detection alone.

Conclusion

This paper has presented an AI-powered e-waste recycling and management platform that integrates automated object detection with a complete, role-aware recycling workflow within a single digital ecosystem. The proposed system employs a custom-trained YOLOv8 model for e-waste classification, combined with Haversine-based collector assignment, OTP-secured handovers, digital wallet transactions, and an administrative monitoring dashboard. Experimental evaluation on a dataset of 19,613 annotated images yielded a Precision of 66.3%, Recall of 60.6%, mAP@0.5 of 62.4%, and mAP@0.5:0.95 of 48.5%. These results reflect moderate and practically viable detection performance, and demonstrate the feasibility of deploying AI-based classification within a real-world recycling platform. Functional testing of the platform modules confirmed reliable performance across collector assignment, OTP verification, wallet transactions, and administrative controls. Several limitations of the current system merit consideration. The training dataset exhibits class imbalance, with certain e-waste categories underrepresented relative to others, which affects per-class recall. Additionally, certain device types share significant visual similarity such as mobile phones and tablets leading to inter-class confusion under ambiguous conditions. The model's performance is also sensitive to image quality, with degraded detection accuracy observed under poor lighting, cluttered backgrounds, and low-resolution inputs. These limitations suggest specific directions for future improvement. Along with classification, the platform facilitates secure and transparent collaboration among customers, collectors, recyclers, and administrators, thereby improving operational efficiency across the recycling lifecycle. Future work will focus on expanding and rebalancing the training dataset, incorporating data augmentation strategies to improve robustness under real-world imaging conditions, exploring model compression for edge deployment, and conducting large-scale user studies to validate usability and reliability. Overall, the proposed system demonstrates the meaningful potential of integrating artificial intelligence with structured digital workflow management to advance sustainable, transparent, and efficient e-waste recycling practices.

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